

Lesson 5 Docker Hardware Interaction and Data Processing

1. Hardware Mounting (Port Binding)

Connect the device to be mounted to the master, and create a udev rule in the host (/etc/udev/rules.d/).

Here is an example of mounting a USB camera. Please note that the example is for reference only and you should create your own udev rule based on your specific device.

1) Press “Ctrl+Alt+T” to open the command line terminal, and enter “docker run -it --device=/dev/video0 debian:latest /bin/bash” in the terminal to start the “debian” container and mount the USB camera.

```
pi@raspberrypi:~ $ docker run -it --device=/dev/video0 debian:latest /bin/bash
```

2) Enter “ls /dev/” in the terminal and press “Enter” to check the device currently mounted in the container. You can see that the USB camera "video0" has been successfully mounted.

```
pi@raspberrypi:~ $ docker run -it --device=/dev/video0 debian:latest /bin/bash
root@80ead1e0089b:/# ls /dev/
console fd full mqueue null ptmx pts random shm stderr stdin stdout tty urandom video0 zero
```

2. GUI Displaying in Docker

Install in the host:

1) Press “Ctrl+Alt+T” to open the command line terminal, and enter “sudo apt-get install tigervnc-standalone-server tigervnc-viewer” in the terminal. Press “Enter” and enter “y” to download.

```
pi@raspberrypi:~$ sudo apt-get install tigervnc-standalone-server tigervnc-viewer
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libfile-readbackwards-perl libfltk-images1.3 tigervnc-common tigervnc-tools xfonts-base
Suggested packages:
  xfonts-100dpi | xfonts-75dpi xfonts-scalable
The following NEW packages will be installed:
  libfile-readbackwards-perl libfltk-images1.3 tigervnc-common tigervnc-standalone-server tigervnc-tools tigervnc-viewer
  xfonts-base
0 upgraded, 7 newly installed, 0 to remove and 108 not upgraded.
Need to get 7,294 kB of archives.
After this operation, 11.9 MB of additional disk space will be used.
Do you want to continue? [Y/n]
```

2) Enter “sudo apt-get install x11-xserver-utils” in the terminal and press “Enter” to download.

```
pi@raspberrypi:~$ sudo apt-get install x11-xserver-utils
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
x11-xserver-utils is already the newest version (7.7+9+b1).
x11-xserver-utils set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 108 not upgraded.
```

3) Enter “xhost +” in the terminal and press “Enter” to download.

```
pi@raspberrypi:~$ xhost +
access control disabled, clients can connect from any host
```

4) Enter “docker run -it --env=“DISPLAY” --env=“QT_X11_NO_MITSHM=1” -v /tmp/.X11-unix:/tmp/.X11-unix debian:latest /bin/bash” in the terminal and press “Enter” to get into the container.

```
pi@raspberrypi:~$ docker run -it --env="DISPLAY" --env="QT_X11_NO_MITSHM=1" -v /tmp/.X11-unix:/tmp/.X11-unix debian:latest /bin/bash
root@2faedbd22176:/#
```

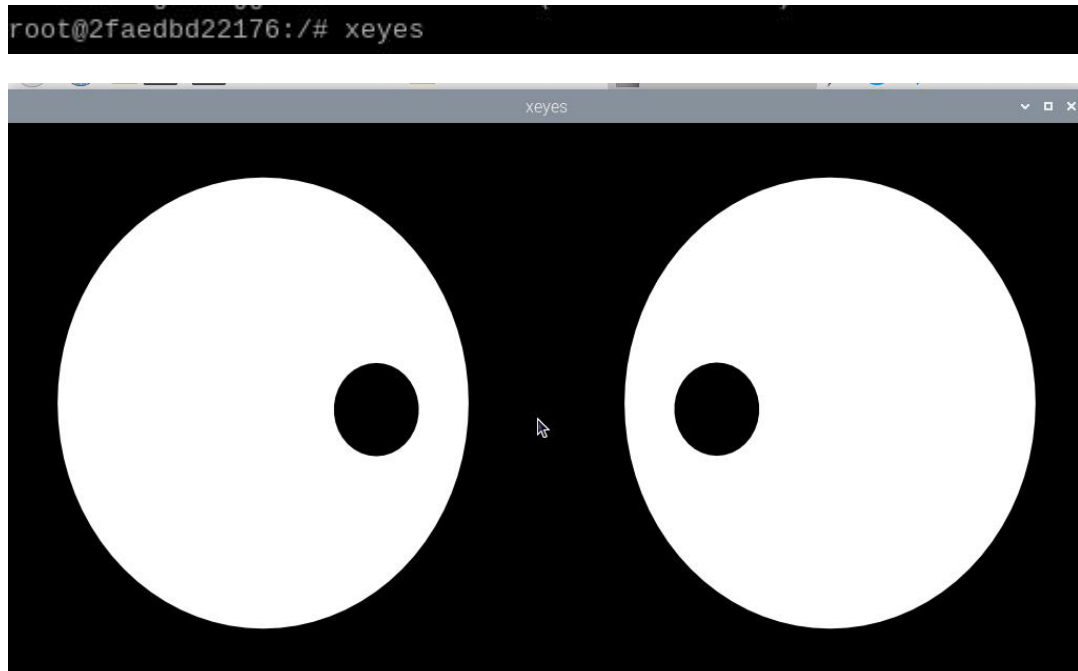
5) Enter “apt-get update” in the terminal and press “Enter” to update the software list.

```
root@2faedbd22176:/# apt-get update
Get:1 http://deb.debian.org/debian bookworm InRelease [151 kB]
Get:2 http://deb.debian.org/debian bookworm-updates InRelease [52.1 kB]
Get:3 http://deb.debian.org/debian-security bookworm-security InRelease [48.0 kB]
Get:4 http://deb.debian.org/debian bookworm/main arm64 Packages [8685 kB]
Get:5 http://deb.debian.org/debian bookworm-updates/main arm64 Packages [12.5 kB]
Get:6 http://deb.debian.org/debian-security bookworm-security/main arm64 Packages [132 kB]
Fetched 9080 kB in 5s (1757 kB/s)
Reading package lists... Done
root@2faedbd22176:/# apt install -y x11-apps
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

6) Enter “apt-get install -y x11-apps” in the terminal and press “Enter” to install the X11 image interface.

```
root@2faedbd22176:/# apt install -y x11-apps
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bsdextrautils fontconfig-config fonts-dejavu-core groff-base libbrotli1 libbsd0 libexpat1 libfontconfig1 libfreetype6
  libgdbm6 libice6 libpipeline1 libpng16-16 libsm6 libuchardet0 libx11-6 libx11-data libx11-xcb1 libxau6 libxaw7
  libxcb-damage0 libxcb-present0 libxcb-randr0 libxcb-xfixes0 libxcb1 libxcursor1 libxdmcp6 libxext6 libxfixes3 libxft2 libxi6 libxkbfile1
  libxmu6 libxmuu1 libxpm4 libxrender1 libxt6 man-db x11-common xbitmaps
```

7) Enter “xeyes” in the terminal and press “Enter”. A pair of eyes will appear, following the movement of the mouse.



3. File Transmission between Docker Container and Host

3.1 Copy File with “cp” Command

3.1.1 Copy File from Container to Host

Command: `docker cp [container id: container path] [target host path]`

1) Enter “`docker run -it debian /bin/bash`” in the terminal to get into the container.

```
pi@raspberrypi:~$ docker run -it debian /bin/bash
root@d2d0e43b41eb:/#
```

2) Enter “`cd`” in the terminal to switch to the user's home directory.

```
root@d2d0e43b41eb:/# cd
```

3) Enter “`ls`” in the terminal to check the directory in the current path. You can see that it is an empty directory.

```
root@d2d0e43b41eb:~# ls
```

4) Enter “`touch test.txt`” in the terminal to create a “txt” file.

```
root@d2d0e43b41eb:~# touch test.txt
```

5) Enter “`pwd`” in the terminal to check the current path.

```
root@d2d0e43b41eb:~# pwd
/root
```

6) Press “`ctrl+P+Q`” to exit the container without stopping it.

7) Enter “`docker cp d2d0e43b41eb:/root/test.txt ~/`” in the terminal to copy the “test.txt” file created earlier to the host.

```
pi@raspberrypi:~$ docker cp d2d0e43b41eb:/root/test.txt ~/
Successfully copied 1.54kB to /home/pi/
```

8) Enter “`cd /home/pi`” in the terminal to access the file path.

```
pi@raspberrypi:~ $ cd /home/pi/
```

9) Enter "ls" in the terminal to see that "test.txt" is already in the path.

```
pi@raspberrypi:~ $ ls
Bookshelf  debian.tar  Dockerfile  Downloads  Pictures  Templates  Videos
debian_1.1.tar  Desktop    Documents   Music      Public    test.txt
```

3.1.2 Copy File from Host to Container

Command: `docker cp [host file path] [container id: container path]`

1) Enter "touch test_1.txt" in the terminal to create a "test_1.txt" file.

```
pi@raspberrypi:~ $ touch test_1.txt
```

2) Enter "docker cp test_1.txt d2d0e43b41eb:/root/" in the terminal to copy the "test_1.txt" file to the "/root/" path of the container "d2d0e43b41eb" (Note: the container "d2d0e43b41eb" needs to keep running).

```
pi@raspberrypi:~ $ docker cp test_1.txt d2d0e43b41eb:/root/
Successfully copied 1.54kB to d2d0e43b41eb:/root/
```

3) Enter "docker exec -it d2d0e43b41eb /bin/bash" in the terminal to enter the container.

```
pi@raspberrypi:~ $ docker exec -it d2d0e43b41eb /bin/bash
```

4) Enter "cd" in the terminal to switch to the user's home directory.

```
root@d2d0e43b41eb:/# cd
```

5) Enter "ls" in the terminal to check the directory in the current path. You can see that the file has been copied from the host to the container.

```
root@d2d0e43b41eb:~# ls
test.txt  test_1.txt
```

4. Data Volume Usage

4.1 Data Volume Introduction

When an application and its running environment are packaged into a container, the container can run along with the application. However, the requirement for the data is that it can be persistent. For example, if a “mysql” is installed in a container which is then deleted, the database would be lost. Therefore, we want containers to be able to share data. If the docker commit is not used to generate a new image and save the data as a part of the image, the data generated by the container will be lost when the container is deleted.

To solve this problem, data volumes can be used to share data between containers and persist the data even after the container is deleted.

Features:

1. Data volumes can be shared or reused between containers.
2. Changes to data volumes take effect immediately.
3. Changes to data volumes are not included in image updates.
4. The life cycle of a data volume continues until no container is using it.

4.2 Data Volume Usage

Command: `docker run -it -v [host absolute path directory: container directory] [image name]`

1) Enter “`docker run -it -v /home/pi/temp:/root/temp debian:latest /bin/bash`” in the terminal to share the “/home/pi/temp” directory on the host with the “/root/temp” directory in the container.

```
pi@raspberrypi:~$ docker run -it -v /home/pi/temp:/root/temp debian:latest /bin/bash
root@46de34b3ffb7:/# ls
```

2) Enter “cd root/temp/” in the terminal to enter the specified directory.

```
root@46de34b3ffb7:/# cd root/temp/
```

3) Enter “touch text_2.txt” in the terminal to create a file named “touch text_2.txt”.

```
root@46de34b3ffb7:~/temp# touch text_2.txt
```

4) Enter “exit” in the terminal to exit the container.

```
root@46de34b3ffb7:~/temp# exit
exit
pi@raspberrypi:~$
```

5) Enter “cd /home/pi/temp/” in the terminal to enter the specified directory.

```
pi@raspberrypi:~$ cd /home/pi/temp/
pi@raspberrypi:~/temp$
```

6) Enter “ls” in the terminal to see that the file created in the container has been shared to the specified directory.

```
pi@raspberrypi:~/temp$ ls
text_2.txt
```